

Toufiq Musah

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Summary

Artificial Intelligence and Computer Vision for Medicine and Biomedical Data Applications. Efficient Representation Learning with Multimodal Data, and Causal Machine Learning Methods for Clinical Reasoning and Decision Support.

Education

University of Pennsylvania

Aug 2025 – Present

M.S.E Data Science

Relevant Courses: Machine Learning for Data Science, Mathematical Foundations for Machine Learning, Computer Systems Programming, Big Data Analytics

Kwame Nkrumah University of Science and Technology (KNUST)

Jan 2021 – Nov 2024

BSc. Biomedical Engineering – First Class Honors

Thesis: An Explainable Artificial Intelligence Framework for Clinical Decision Support Systems in Stroke Triaging

Relevant Courses: Research Methods, Biomechanics, Medical Imaging, Biosignal Processing and Analysis, Probability and Statistics, Linear Algebra, Multivariable Calculus, C/C++, Biomaterials, Bioinstrumentation,

Relevant Professional Experience

Research Assistant and Engineering

Kumasi, Ghana

Oct 2024 – Oct 2025

Kumasi Centre for Collaborative Research in Tropical Medicine (KCCR)

- Developed a context-specific Large Language Model (LLM) application utilizing Retrieval Augmented Generation and Efficient Fine-tuning, to function as a tool for the management of Gestational Diabetes Mellitus, improving information access and self-management for expectant mothers.
- Conducted interdisciplinary research and collaborated on systematic reviews in Artificial Intelligence/Machine Learning and Digital Health in the Global Health & Infectious Diseases (GHID) Research Group.

Research Assistant, Machine Learning Intern

Kumasi, Ghana

Oct 2023 – Feb 2024

Responsible Artificial Intelligence Lab (RAIL)

- Conducted extensive research across multiple domains of machine learning/deep learning, including Generative Adversarial Networks (GANs) and Machine Learning Applications in Biomedical Engineering.
- Built Generative Models for Computed Tomography (CT) denoising and upscaling, and CT-MRI translation and style-transfer.

Research and Academic Work

Peer-Reviewed Workshop Publications

[C.1] Musah Toufiq, 'Large Kernel MedNeXt for Breast Tumor Segmentation and Self-normalizing Network for pCR Classification in Magnetic Resonance Images'. Artificial Intelligence and Imaging for Diagnostic and Treatment Challenges in Breast Care. Deep-Breath 2025. Lecture Notes in Computer Science, vol 16142. Springer, Cham. https://doi.org/10.1007/978-3-032-05559-0_8. [ArXiv] [Springer]

[C.2] Musah Toufiq, Adjei Prince Ebenezer and Otoo, Kojo Obed, 'Automated Segmentation of Ischemic Stroke Lesions in Non-Contrast Computed Tomography Images for Enhanced Treatment and Prognosis'. Medical Information Computing. MIMA EMERGE 2024. Communications in Computer and Information Science, vol 2240. Springer, Cham. https://doi.org/10.1007/978-3-031-79103-1_8. [ArXiv] [Springer]

Accepted Workshop Papers | Preprints Awaiting Release

[C.3] Musah Toufiq, Kalaiwo Chinasa, Akram Maimoona, Abdulai Ubaida Napari, Adewole Maruf, Dako Farouk, Emegoakor Adaobi Chiazor, Anazodo Udunna C, Adjei Prince Ebenezer and Raymond, Confidence. 'Towards Trustworthy Breast Tumor Segmentation in Ultrasound using Monte Carlo Dropout and Deep Ensembles for Epistemic Uncertainty Estimation'. Accepted at: Medical Image Computing in Resource Constrained Settings Workshop. 28th International MICCAI Conference, 2025. [ArXiv]

[C.4] Ankomah Claudia Takyi, Ayivor Livingstone Eli, Nyame Ireneaus, Wambo Leslie, Bonsu, Patrick Yeboah, Iorumbur Aondona Moses, Confidence Raymond and Musah Toufiq. 'How We Won BraTS-SSA 2025: Brain Tumor Segmentation in the Sub-Saharan African Population Using Segmentation-Aware Data Augmentation and Model Ensembling'. Accepted at: Brain Tumor Segmentation, and Cross-Modality Domain Adaptation for Medical Image Segmentation (BraTS). 28th International MICCAI Conference, 2025. [ArXiv]

[C.5] Musah Toufiq, Nkwam Philip, Sharma Ajay. 'Automated Pediatric Brain Hippocampal and Basal Ganglia Segmentation in Ultra-Low Field Magnetic Resonance Images'. Accepted at: MICCAI Challenge on Low Field Pediatric Brain Magnetic Resonance Image Segmentation and Quality Assurance, 2025

Selected Talks & Orals

Automated Medical Image Segmentation: Toward Trustworthy Clinical Tools, COMPASS, 2025 [C.1,3,4]
Towards Trustworthy Breast Tumor Segmentation, MIRASOL Workshop, MICCAI Conference, 2025 [C.3]
Large Kernel MedNeXt for Breast Tumor Segmentation, MIDL Young Researcher Showcase, 2025 [C.1]
Optimisation Techniques in Machine Learning, Ghana Data Science Summit, Tutorial, 2025

Relevant Projects

Multi-Modal Data Integration for Head and Neck Tumor Prognostication

- Developed a multimodal learning framework to predict 5-year survivability and recurrence in head and neck tumor cases using whole-slide image features, structured clinical data, laboratory analytes, surgical notes.
- Generated tiny ClinicalBERT embeddings from free-text notes and fused modalities via a low-rank multimodal bilinear fusion, achieving an AUC of 0.75 across prognostic tasks.

Vision-Language Modeling for Surgical Scene Understanding & Panoptic Segmentation

- Surgical scene understanding using surgical video and panoptic segmentations as multimodal inputs.
- Trained S²-Scaled SwinUNETR for panoptic segmentation across 21 classes, enabling precise spatial localization.
- Fine-tuned MedGemma with QLoRA for scene graph generation and automated detection of key surgical events.

Collaborative AI Resources for Expectant Mothers (C.A.R.E)

- Developed an accessible LLM-powered assistant using Retrieval-Augmented Generation to support expectant mothers managing gestational diabetes.
- Deployed the system through WhatsApp to maximize accessibility in resource-limited settings.

Generative AI for Medical Imaging Enhancement and Data Augmentation

- Designed a super-resolution and denoising pipeline to generate high-quality head CT scans from low-resolution images to reduce effective radiation exposure.
- Implemented a Deep Convolutional Generative Adversarial Network for synthetic medical image generation for unsupervised pre-training and efficient downstream finetuning.

Awards and Acknowledgments

- Best Oral Presentation Award - MICCAI Meets Africa Workshop, 2024 [C.2]
- Brain Tumor Segmentation Challenge (BraTS-SSA Task), 2025 – First Place [C.4]
- Best Poster Presentation Award - Ghana Data Science Summit, IndabaX, 2024
- Academic Excellence Award – Provost List 2021, 2022, 2023
- MasterCard Healthcare Entrepreneurship Grant
- MICCAI Education Challenge Finalist, 2024
- MICCAI Travel Award, 2024, 2025

Volunteering and Community Activities

Team Coordinator, Sprint AI Training for Medical Image Knowledge Translation

Developing training materials, and coordinating the [Sprint AI training program](#), focused on building capacity among emerging researchers and clinicians in medical image computing and deep learning applications.

Technical Committee, Breast Ultrasound Segmentation and Classification Challenge

Contributed to the organization of the [PRECISE-Abreast Challenge, 2025](#). Implemented evaluation pipelines for submissions, ensuring fairness and reproducibility of the assessment process.

Course Facilitator, Women in Engineering - SheCodes Club

Teaching Python programming and introductory machine learning classes for the Women in Engineering Society, creating an inclusive learning environment to help members gain tech skills and build their confidence in STEM.

MONAI Ultrasound Working Group Volunteering in the MONAI Ultrasound Working Group, working towards the development of a foundational model for general structure segmentations in 2D and 3D Ultrasound

IndabaX, Ghana Data Science Summit Contributing to the organization of Ghana Data Science Summit. Worked with various committees to promote the event. Hosted an engaging tutorial session at the event.

Blog Posts, Tutorials

A Hitchhiker's Guide to 3D Medical Image Processing – [Medium Article](#)

Introduction to TorchIO for 3D MRI Processing – [Preprocessing \(Part 1\)](#) - [Augmentation & Dataloaders \(Part 2\)](#)

End-to-End Deep Learning Tutorial for Image Classification: Pneumonia Detection – [Colab Notebook](#)

Skills

Languages: Python, LaTeX, C/C++, MATLAB | **Libraries:** PyTorch, TensorFlow, Scikit-Learn

Software: Solidworks, ANSYS FEA, Unity Game Engine, 3D Slicer, NiBabel